Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Which of the following diagrams is correctly related to the Parke's process for the extraction of Ag form pb-Ag alloy ?

(A) $Ag - pb \xrightarrow[(Cupellation)]{Heating\ with\ O_2} pure\ Ag$

(B) $Ag - pb \overset{\text{Distllation}}{\longrightarrow} Pb_{(g)}$ and $Ag_{(s)}$

(C) $Ag - pb \xrightarrow[\text{Zn}]{\text{molted}} Zn - pb$ layer and Ag layer

(D) $Ag - pb \xrightarrow[\text{Zn}]{\text{molted}} Zn - Ag$ layer and pb layer

Q.2

Among the following statements, the incorrect one is :

(A) cassiterite ore of tin contains the impurities of Wol-framite which are separated by electromagnetic separator.

(B) tin metal is obtained by the carbon reduction of black tin.

(C) in the extraction of lead from galena, the roasting and self-reduction are carried out in the same furnance at different temperature.

(D) none of these

Q.3

During the electrolysis of carnallite, $\mathrm{MgCl_2}$ is decomposed and not KCl. This is because of :

(A) lower decomposition voltage of $\mathrm{MgCl_2}$ than that of KCl

(B) reverse reaction $\mathrm{MgCl_2 + 2\ K \rightarrow Mg + 2KCl}$ If KCl is decomposed under other experimental condition.

(C) both (A) and (B)

(D) none of the above

Q.4

Main source of lead is galena (PbS). It is converted to $\mathbf{Pb}$ by :

(A) :PbS $\xrightarrow[\Delta]{air}$ PbO + SO_2
PbO $\xrightarrow{C}$ Pb + CO_2

(B) :PbS $\xrightarrow[\Delta]{air}$ PbO + PbS
PbO + PbS $\rightarrow$ Pb + SO_2

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Self-reduction process is :

(A) A

(B) B

(C) both

(D) none

Q.5

S_1 : Cuprite, Limonite and Zincite are oxide ores
S_2 : Magnesite and carnallite are magnesium ores
S_3 : Chalcocyte and azurite are ores of copper
S_4 : Feldspar and mica minerals contain aluminium.

(A) T T T T

(B) F T F F

(C) F F T T

(D) T T F T

Q.6

When the ore, haematite is burnt in air with coke around $2000^{\circ}\mathrm{C}$ along with lime, the process not only produces steel but also produces a slag that is useful in making building materials. The slag is

(A) $\mathrm{NaSiO_3}$

(B) $\mathrm{CaSiO_3}$

(C) $\mathrm{SiO_2}$

(D) $\mathrm{CaCO_3}$

Q.7

Which of the following statement is correct?

(A) In Hall-Heroult process, the electrolyte used is a molten mixture of ahumina, sodium hydroxide and cryolite.

(B) Lead is extracted form its chief ore by both carbon reduction and self reduction.

(C) Tin is extracted from its chief ore by carbon monoxide reduction.

(D) Siderite, cassiterite and argentite are carbonate ores.

Q.8

For extraction of sodium from NaCl, the electrolytic mixture $\mathrm{NaCl} + \mathrm{Na_3AlF_6} + \mathrm{CaCl_2}$ is used. During extraction process, only sodium is deposited on cathode but K and C a do not because

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(A) Na is more reactive than K and Ca

(B) Na is less reactive than K and Ca

(C) $NaCl$ is less stable than Na_3AlF_6 and $CaCl_2$

(D) the discharge potential of Na^+ is less than that of K^+ and Ca^{+2} ions

Q.9

If Bauxite consists of SiO_2 as impurity, this process is employed -

(A) Hall's process

(B) Baeyer's process

(C) Hoope's process

(D) serpeck's process

Q.10

Formation of metallic copper from the sulphide ore in the commercial thermo-metallurgical process essentially involves which one of the following reaction :

(A) $Cu_2\ S + 3/2O_2 \longrightarrow Cu_2O + SO_2; CuO + C \longrightarrow Cu^2 + CO$

(B) $Cu_2\ S + 3/2O_2 \longrightarrow Cu_2O + SO_2; 2Cu_2O + Cu_2\ S \longrightarrow 6Cu + SO_2$

(C) $Cu_2\ S + 2O_2 \longrightarrow CuSO_4; CuSO_4 + Cu_2\ S \longrightarrow 3Cu + 2SO_2$

(D) $Cu_2\ S + 3/2O_2 \longrightarrow Cu_2O + {}^2SO_2; Cu_2O + CO \longrightarrow 2Cu + CO_2$

Q.11

$Ag_2\ S + NaCN \rightarrow (A), (A) + Zn \rightarrow (B) + Ag$

(A) and (B) are respectively

(A) $Na_2\left[Zn(CN)_4\right], Na_2\left[Ag(CN)_4\right]$

(B) $Na\left[Ag(CN)_2\right], Na_2\left[Zn(CN)_4\right]$

(C) $Na\left[Ag(CN)_2\right], Na\left[Zn(CN)_3\right]$

(D) $Na_3\left[Ag(CN)_3\right], Na_2\left[Zn(CN)_4\right]$

Q.12

Which of the following is not used for extraction of Aluminium -

(A) Van arkel process

(B) Serpeck's process

(C) Baeyer's process

(D) Hall-Heroult's process

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Q.13

Heating an ore in the abesence of air below its melting point is called:

(A) leaching

(B) roasting

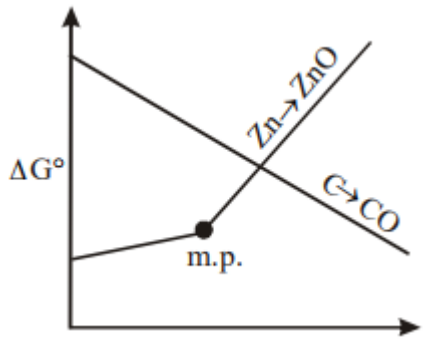
(C) smelting

(D) calcination

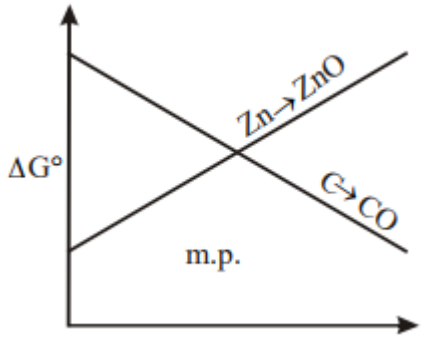
Q.14

Metallic zinc is collected in the form of vapour by the carbon reduction process, then the incorrect diagram for this phenomenon is/are -

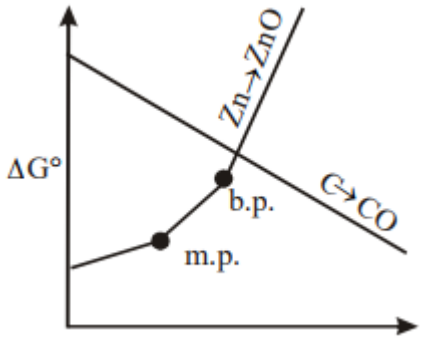
(A)



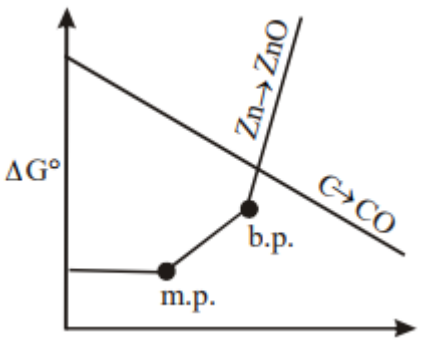
(B)



(C)



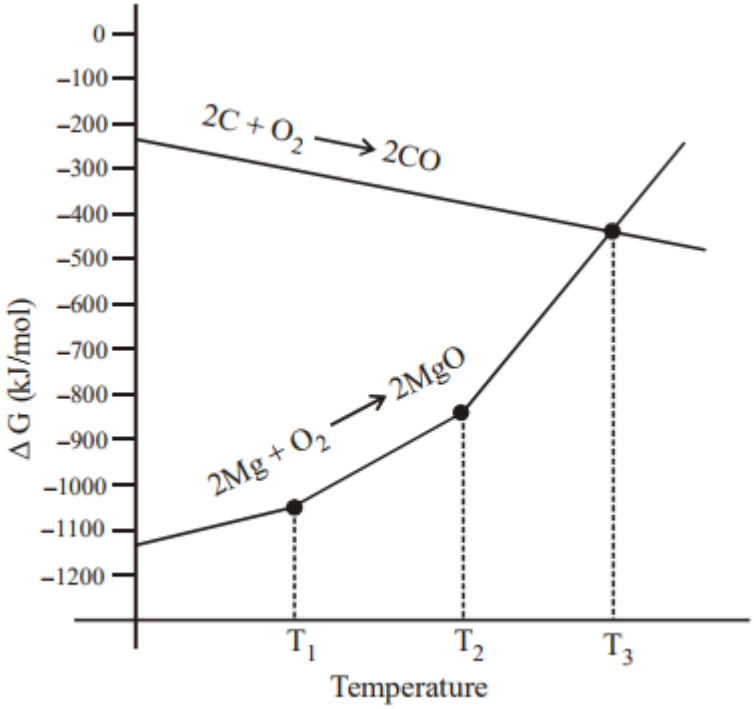
(D)



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Q.15

Correct statements about the plot is/are —



(A) T_1 and T_2 are melting point and boiling point of Mg respectively

(B) T_1 and T_2 are melting point and boiling point of MgO respectively

(C) Reduction of MgO by coke is possible above T_3

(D) Mg can be extracted from gaseous products by rapid cooling

Q.16

Choose the correct options -

(A) The reduction of a metal oxide is easier if the metal formed is in liquid state at the temperature of reduction

(B) The value of $\Delta_f G^o$ for formation of Cr_2O_3 is $-540\ kJ\ mol^{-1}$ and that of Al_2O_3 is $-827\ kJ\ mol^{-1}$. The reduction of Cr_2O_3 is possible with Al.

(C) Carbon monoxide is more effective reducing agent than carbon below $983\ K$ but above this temperature, the reverse is true.

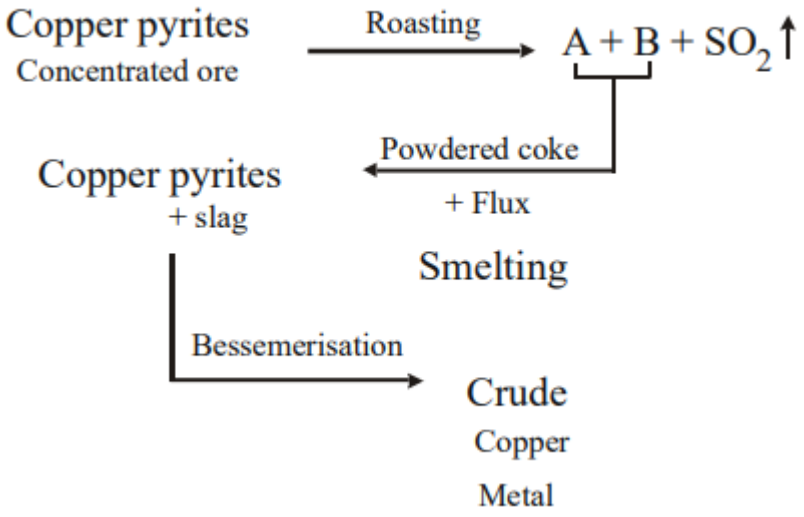
(D) None of these

Q.17

Passage

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Copper is extracted from its ore copper pyrites through the scheme given below.



Question

The copper ore (copper pyrites) is concentrated by:

(A) Leaching

(B) Froath-floatation process

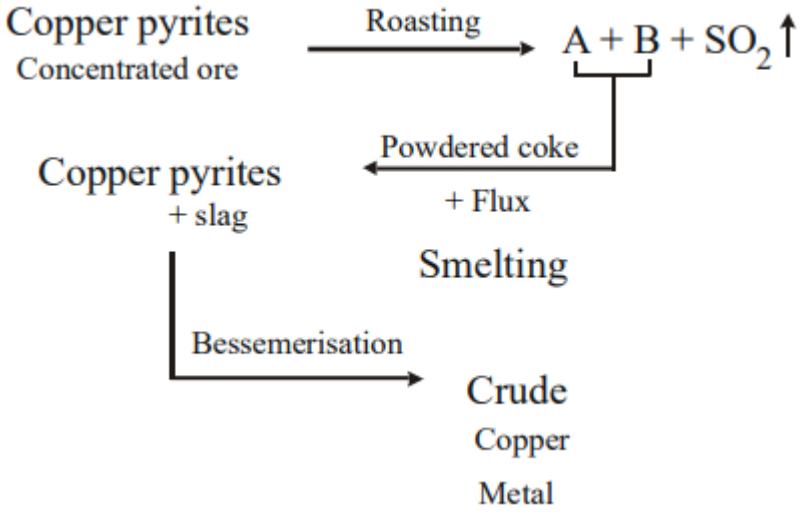
(C) Levigation process

(D) Magnetic separation

Q.18

Passage

Copper is extracted from its ore copper pyrites through the scheme given below.



Question

The formula of the copper pyrites is:

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(A) Cu_2S

(B) Cu_2FeS_2

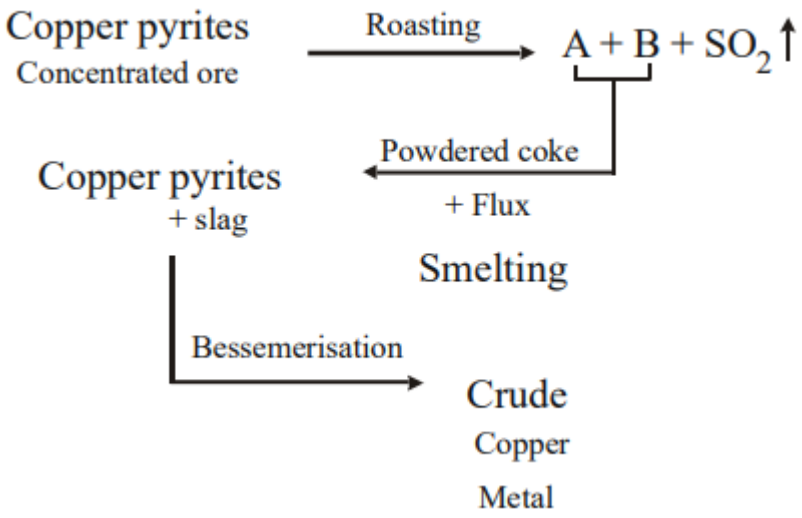
(C) $CuFeS_2$

(D) $Cu(OH)_2 \cdot CuCO_3$

Q.19

Passage

Copper is extracted from its ore copper pyrites through the scheme given below.



Question

The compounds A and B are:

(A) CuS ad FeS

(B) Cu_2S and FeS

(C) Cu_2S and FeO

(D) Cu_2S and Fe_2O_3

Q.20

Passage

Copper is the most noble of the first row transition metals and occurs in small deposits in several countries, Ores of copper include chalcanthite $(CuSO_4.5H_2O)$, atacamite $(Cu_2Cl(OH)_3)$, cuprite (Cu_2O), copper glance (Cu_2S) and malachite $(Cu_2(OH)_2CO_3)$. However, 80% of the world copper production comes from the ore chalcopyrite $(CuFeS_2)$. The extraction of copper from chalcopyrite involves partial roasting, removal of iron and self-reduction.

Question

Partial roasting of Chalcopyrite produces -

(A) Cu_2S and FeO

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(B) Cu_2O and FeO

(C) CuS and Fe_2O_2

(D) Cu_2O and Fe_2O_2

Answers & Solutions

Q.1 Answer:

$Ag - pb \xrightarrow[Zn]{\text{molted}} Zn - Ag$ layer and pb layer

Solution:

Molted Zn is not soluble in molted Pb.
Ag is soluble in molted Pb and in molted Zn
Solubility of Ag is higher in Zn layer as compared to Pb layer.
Molted Zn is lighter as compared to Pb

Q.2 Answer:

none of these

Solution:

(A) Cassiterite contains iron and manganese tungstate, called as Wol-framite. Theyhave magnetic properties.
(B) Purified ore that contains $70\%SnO_2$ is called black tin.
$SnO_2 + 2C \rightarrow Sn + 2CO$ **(carbon reduction)**
(C) Roasting is done at moderate temperature in presence of air in reverberatory furnace. Then air supply is stopped and temperature is increased to melt the mass when self-reduction takes place.

Q.3 Answer:

both (A) and (B)

Solution:

$MgCl_2$ **has lower decomposition voltage than that of** KCl
$MgCl_2 + K \rightarrow 2KCl \rightarrow$ **Reverse reaction**

Q.4 Answer:

B

Solution:

PbS **reduces** PbO **to** Pb **and** SO_2 **is liberated. This is called as self-reduction.**

Q.5 Answer:

T T T T

Solution:

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$S_1 : Cu_2O, Fe_2O_3 \cdot 3H_2O, ZnO$
$S_2 : MgCO_3, KCl \cdot MgCl_2 \cdot 6H_2O$
$S_3 : Cu_2\ S, 3CuCO_3 \cdot Cu(OH)_2;$
S_4 : **Both contain aluminium.**

Q.6 Answer:

$CaSiO_3$

Solution:

$C + O_2 \longrightarrow CO_2$
$CO_2 + C \longrightarrow CO; Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$
$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$
$CaCO_3 \xrightarrow{\Delta} SiO_2 \longrightarrow CaSiO_3$ **(slag)**

Q.7 Answer:

Lead is extracted form its chief ore by both carbon reduction and self reduction.

Solution:

(A) Molten mixture of alumina and cryolite
(B) $2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$
$PbS + 2PbO \longrightarrow 3\ Pb + SO_2$ **(self reduction)**
$PbO + C \longrightarrow Pb + CO$ **(carbon reduction)**
(C) $SnO_2 + C \longrightarrow Sn + CO$ **(carbon reduction)**
(D) siderite $= FeCO_3$**; cassiterite** $= SnO_2$ **and argentite** $= Ag_2\ S.$

Q.8 Answer:

the discharge potential of Na^+ is less than that of K^+ and Ca^{+2} ions

Solution:

Here K **and** Ca **do not deposited because the discharge potential of** Na^+ **is less than that of** K^+ **and** Ca^{2+} **ions i.e , Discharge potential order** $Na < Ca < K$

Q.9 Answer:

serpeck's process

Solution:

Bayer \& Hall's process are used when Fe_2O_3 **is major impurity. Hoope's process is a refining process.**

Q.10 Answer:

$Cu_2\ S + 3/2O_2 \longrightarrow Cu_2O + SO_2; 2Cu_2O + Cu_2\ S \longrightarrow 6Cu + SO_2$

Solution:

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Self reduction process

$$Cu_2S + \frac{3}{2}O_2 \rightarrow Cu2O + SO_2$$

$$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$$

Q.11 Answer:

$Na[Ag(CN)_2], Na_2[Zn(CN)_4]$

Solution:

$$Ag_2S + 4NaCN \longrightarrow 2Na[Ag(CN)_2] + Na_2S$$

$$2Na[Ag(CN)_2] + Zn \longrightarrow Na_2[Zn(CN)_4] + 2Ag$$

Q.12 Answer:

Van arkel process

Solution:

Van Arkel process is one the refining process to get very pure metal. In this process the impure metal is converted into volatile Iodides, which are again dissociated to get pure metal.

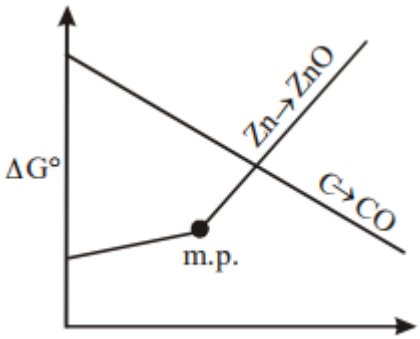
Q.13 Answer:

calcination

Solution:

Roasting is heating in presence of air.

Q.14 Answer:



Solution:

(A) Represents extraction of Zn in its molten state

(B) Unable to represent the phase of extracted zinc.

(C) Represents that zinc will be extracted in the form of vapours

$$Zn(g) + \tfrac{1}{2}O_2(g) \longrightarrow ZnO(g)$$

$$ZnO(g) + C(s) \xrightarrow[\text{B.P. of Zn}]{\text{above}} Zn(g) + CO\uparrow$$

(D) Represents wrong Ellingham diagram for formation of ZnO, as

$$Zn(s) + \tfrac{1}{2}O_2(g) \xrightarrow[400^\circ C]{\text{below}} ZnO(s)$$

$\Rightarrow$ **Decreases in entropy in the forward direction**

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At $420^{\circ}C : Zn(\ell) + \frac{1}{2}O_2(g) \longrightarrow ZnO(g)$

$\Rightarrow$ **Decreases in entropy in forward direction.**

Q.15 Answer:

T_1 and T_2 are melting point and boiling point of Mg respectively

Solution:

(A) At T_1 **and** T_2 **the value of** ΔG_f **for** MgO **becomes more and more positive due to decrease in entropy in the formation of** MgO.

$$2Mg(\ell) + O_2(g) \xrightarrow{T_1} MgO(s)$$

$$2Mg(g) + O_2(g) \xrightarrow{T_2} 2MgO(s)$$

(C) Beyond T_3, ΔG_f **of** CO **is more negative than that of** MgO

(D) $MgO(s) + C(s) \overset{T_3}{\rightleftharpoons} \underbrace{Mg(g) + CO\uparrow}_{\text{gaseous product}}$

$\downarrow$ Rapid cooling

$Mg(s)$

Q.16 Answer:

The reduction of a metal oxide is easier if the metal formed is in liquid state at the temperature of reduction

Solution:

(A) The entropy is higher if the metal is in liquid state than when it is in solid state. The value of entropy change (ΔS) **of the reduction process is more on +ve side when the metal formed is in liquid state and the metal oxide being reduced is in solid state. Thus the value of** ΔG° **becomes more on negative side and the reduction becomes easier.**

(B) Reduction of Cr_2O_3 **with** Al **can be described by the equation**

$$Cr_2O_3(s) + 2Al(s) \xrightarrow{\text{heat}} 2Cr(s) + Al_2O_3(s)$$

Then, $\Delta_r G^o = \Sigma \Delta_f G^{\circ}$ **(products)** $- \Sigma \Delta_f G^{\circ}$ **(reactants)**

$$\Delta_r G^o = (\Delta_f G^{\circ}(Al_2O_3) + 2\Delta_f G^{\circ}(Cr) - (\Delta_f G^{\circ}(Cr_2O_3) + 2\Delta_f G^{\circ}(Al))$$

$$\Delta_r G^{\circ} = (-827 \text{ kJ mol}^{-1} + 0) - (-540 \text{ kJ mol}^{-1} + 0) = -287 \text{ kJ mol}^{-1}$$

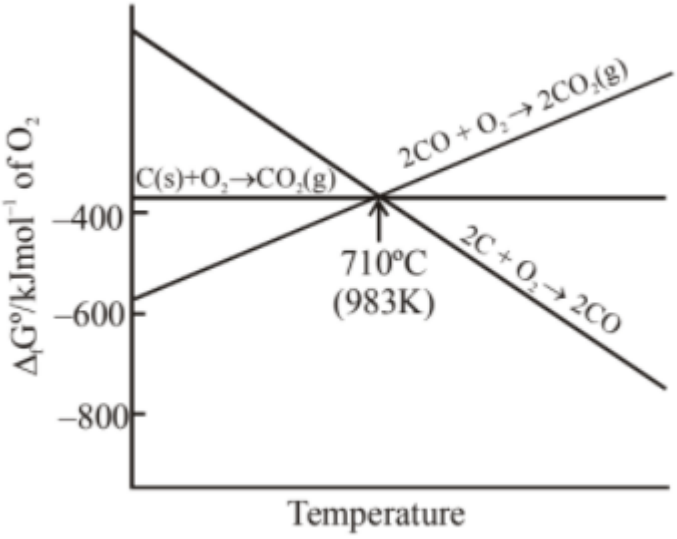
Since $\Delta_r G^{\circ}$ **for the reduction of** Cr_2O_3 **with aluminium is negative, hence the reduction of** Cr_2O_3 **with aluminium is possible.**

(C) The Ellingham diagrams for the formation of CO_2 **and** CO **from** C **and oxygen are shown.**

From the diagram, it is clear that below $710^{\circ}C(983\ K)$, CO **curve lies below the carbon curve, whereas above** $983\ K$, CO **oxidation curve lies above the carbon oxidation curve. Thus,** CO **is more effective**

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reducing agent than carbon below $983\ \mathrm{K}$, and above this temperature the reverse is true.



Q.17 Answer:

Froath-floatation process

Solution:

Since copper pvrites is a sulphide ore it is concentrated bv using froth flotation process.

Q.18 Answer:

$\mathrm{CuFeS_2}$

Solution:

The formula of copper pyrites is $\mathrm{CuFeS_2}$

Q.19 Answer:

$\mathrm{Cu_2\ S}$ and FeS

Solution:

$$\mathrm{CuFeS_2} \xrightarrow{\Delta} \underset{(A)}{\mathrm{Cu_2}} + \underset{(B)}{\mathrm{FeS}}$$

Q.20 Answer:

CuS and $\mathrm{Fe_2O_2}$

Solution:

$$2\mathrm{CuFeS_2} + 4\mathrm{O_2} \longrightarrow \mathrm{Cu_2S} + 2\mathrm{FeO} + 3\mathrm{SO_2}$$

Iron is removed in the form of slag of $\mathrm{FeSiO_3}$

$$\mathrm{FeO} + \mathrm{SiO_2} \longrightarrow \mathrm{FeSiO_3}$$

$\mathrm{S^{2-}}$ **acts as reducing species in self reduction reaction** $2\mathrm{Cu_2O} + \mathrm{Cu_2S} \longrightarrow 6\mathrm{Cu} + \mathrm{SO_2}$

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